

EXPENSE TRACKER APPLICATION

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WHAT IS THE EXPENSE TRACKER?

Personal expense tracking
system

Add and manage expenses

Analyze spending habits

Interactive dashboard

Persistent data storage

FEATURES IMPLEMENTED

- ✓ Add expenses with category, amount, date, and description
- ✓ Spending breakdown by category
- ✓ Monthly spending trend analysis
- ✓ Total expenses and expense count
- ✓ Highest and lowest spending categories
- ✓ Delete expense entries
- ✓ CSV-based persistent storage
- ✓ Auto-seeded with sample expenses

PROJECT ARCHITECTURE

Single-process Java web application



Uses Java built-in HTTP server



No Spring Boot



No Maven or Gradle



Lightweight deployment

DATA STORAGE

In-memory ArrayList

CSV file persistence

Auto-save on updates

Auto-load on startup

No database required

Data Visualization



Doughnut chart → Spending by category



Bar chart → Monthly spending trend



Expense statistics

REST API Endpoints

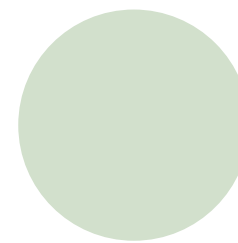
Backend API Design

Method	Endpoint	Purpose
GET	/api/expenses	Retrieve expenses
POST	/api/expenses	Add expense
DELETE	/api/expenses/{id}	Delete expense
GET	/api/stats	Retrieve analytics

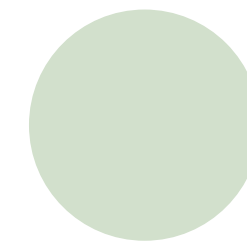
Project Structure

```
expense-tracker/  
├─ run.sh  
├─ data/  
│   └─ expenses.csv  
└─ src/main/java/com/expensetracker/  
    ├─ Expense.java  
    ├─ ExpenseStore.java  
    └─ ExpenseTrackerApp.java
```

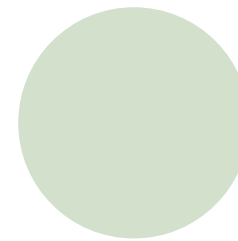
Running the Application: How To Run



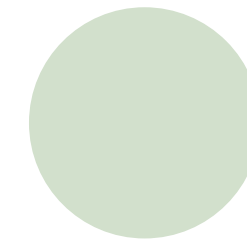
Install Java JDK 25+



Compile Java source files



Run the application



Open browser at localhost:8080

TESTING & VERIFICATION

Seeded with 15 sample expenses

Tested adding expenses

Tested deleting expenses

Verified chart rendering

Verified CSV persistence

CHALLENGES AND SOLUTION

No database allowed
Real-time UI updates
Maintaining
persistence

DESIGN APPROACH

Expense.java → Data
model

ExpenseStore.java →
Storage & business logic

ExpenseTrackerApp.j
ava → Server, API,
and frontend

Conclusion

✓ Lightweight
Java web app

✓ Interactive
analytics
dashboard

✓ Clean
architecture

✓ Persistent
expense tracking

THANK YOU |